Water Quality

ANSWER KEY

Rustin Invitational 2020

Score: 140 pts possible

Record all answers in this packet. Answers recorded in question packet will not be scored.

Part 1: 42 pts

Part 2: 42 pts

Part 3: 42 pts

Part 4: 14 pts

Total: 140 pts

Part 1: Marine and Estuary Ecology - 42 pts (1 pt each unless otherwise noted)

1. abiotic	11. aphotic	21. producer	31. Chesapeake
2. biotic	12. abyssal	22. primary consumer	32. Barrier islands
3. Salt wedge	13. nertic	23. secondary consumer	33. North Carolinae
4. fjord	14. intertidal	24. 10%	34. Salt marshes
5. phytoplankton	15. pelagic	25. g/m^2	35. Mudflats
6. zooplankton	16. benthic	26. $c/m^2/day$	36. submerged aquatic vegetation
7. Slightly Stratified or Partially Mixed Estuary	17. N_2	27. $kg/m^2/yr$	37 in shallow coastal waters
8. Vertically Stratified or Well Mixed Estuaries	18. NH_4^+	28. See below	38. very shallow, light penetrates to bottom, bay grasses are prolific
9. oceanic	19. NO_2^-	29. biological magnification	
10. photic	20. NO_3^-	30. St. Lawernce	

28. Draw an energy pyramid that shows five trophic levels and includes the killer whale. (3 pts) (0.5 for correct diagram, 0.5 for each correctly placed organism)

Must be a 5 level up-ward pointing triangle

5th level: Killer whale
4th level: leopard seal or wendell seal
3rd level: adelie penguin or small fishes or crabeater seal or emperor penguin or large fishes
2nd level: krill or copepods
1st level: diatoms or Phytoplankton

Part 2: Coral Reef Macroflora and Fauna- 42 pts (1 pt each unless otherwise noted)

Number	Organism Common Name	Range (Global, Indo-Pacific, or Atlantic)
1	banded coral shrimp	global
2	humphead wrasse	Indo-Pacific
3	crown of thorns starfish	global
4	Pencil urchin	global
5	Butterfly fish	global
6	flamingo tongue snail	Atlantic
7	barramundi cod	Indo-Pacific
8	Bumphead parrotfish	Indo-Pacific
9	Hard coral	global
10	grouper	global
11	giant clam	Indo-Pacific
12	gorgonia	Atlantic
13	moray eel	global
14	Fleshy algae	global
15	long-spined black sea urchin	global
16	Parrotfish	global
17	Snapper	global
18	sponge	global
19	triton	global
20	nassau grouper	Atlantic
21	sea cucumber	Indo-Pacific
22	Butterfly fish	global
23	Parrotfish	global
24	lobster	global

25 mutualism
26 protozoa
27 C
28 A
29 B
30 E
31 indicator
32 Reef check
33 citizen
34 crown of thorns starfish
35 cleaning other organisms
36 browses on soft coral
37 does not build hard structures
38 filter feeding
39 they eat star-fish
40 encourages the growth of disease causing microorganisms that can kill coral
41 they forage on the agal mat, preventing fleshy algae from overcoming coral
42 by ingesting and passing sand through their digestive system, they dissolve $CaCO_3$, making it available for reef building by hard coral polyps

Part 3: Water Monitoring and Analysis – 42 pts

1. 3.5%	4. $Ca^{2+} + CO_3^{2-} \leftrightarrow CaCO_3$	7. Greater than 4	10. 10
2. 35 ppt	5. 1	8. 8	11. nephalometric turbidity units
3. 35 g/L	6. Low latitudes, near equator	9. 7	12. 5 ppm

Complete the table. (1pt per cell)

Type of pollution	Definition	Symptoms	Effect	Source
Biodegradable waste	13. sewage and animal waste	17. decreasing numbers of fish and other aquatic life; increasing numbers of bacteria	21. decreased oxygen levels; loss of biodiversity	25. run-off or improperly treated effluent; combined sewage overflow
Nutrient pollution	14. nitrates and phosphates	18. green, cloudy, slimy water	22. Algae blooms, eutrophication	26. fertilizer run-off and wastewater treatment
Thermal pollution	15. increased water temperatures	19. water is warmer than normal	23. decreased oxygen levels; loss of biodiversity	27. industrial run-off; wastewater treatment; global warming
Sedimentation	16. suspended particles settling out of water	20. Cloudy water	24. warms up water; toxins accumulate	28. run-off from construction sites, farming and live-stock operations, logging, flooding, city run-off, dams

25. Describe the process of reef bleaching. (4 pts)
1pt – coral get their color from symbiotic zooxanthellae that live within coral tissue
1pt - temp increases by 1 degree C (2 degrees F)
1pt – stressed coral expels zooxanthellae
1pt – coral losses color, may starve

26. Describe the process that produces a dead zone. (4 pts)
1pt – nutrient pollution occurs due to fertilizer run-off; eutrophication
1pt – algal blooms occur
1pt – algae die and decompose
1pt – bacteria deplete oxygen as they decompose the dead algae

Complete the table. 2 pts each cell

Term	Definition
Stenohaline	27. able to tolerate only a narrow range of salinity
Euryhaline	28. Able to adapt to a wide range of salinities
Osmoregulation	29. the maintenance of constant osmotic pressure in the fluids of an organism by the control of water and salt concentrations.

Water Quality

Part 4: Salinometer Testing

Rustin Invitational 2020

Team Number:

School:

Participants:

Score: _____/14

Does the team have a salinometer? Yes or No (7 pts)

Measured salinity of sample: _____ (7 pts)

Full credit given for answers within 1% of actual salinity.
Zero points awarded for answers that are not within 1% of the actual salinity